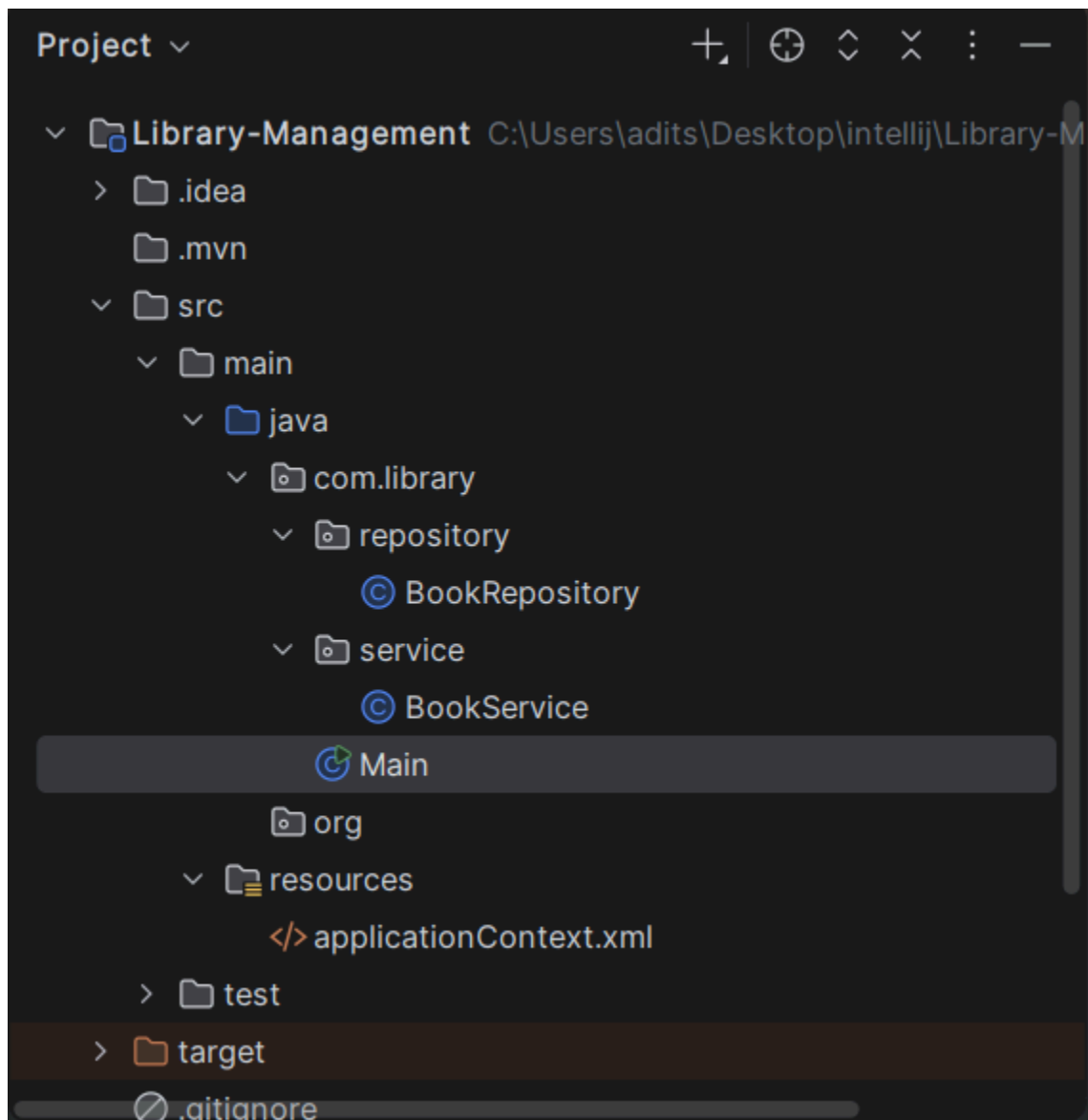


Exercise 7: Implementing Constructor and Setter Injection

File Structure:



Code:

BookRepository.java:

```
package com.library.repository;
```

```

public class BookRepository {

    public void showRepository() {
        System.out.println("Book Repository is working.");
    }

}

```

BoorSerice.java

```

package com.library.service;

import com.library.repository.BookRepository;

public class BookService {

    private BookRepository bookRepository;

    // Constructor Injection
    public BookService(BookRepository bookRepository) {
        this.bookRepository = bookRepository;
        System.out.println("Constructor Injection Done");
    }

    // Setter Injection
    public void setBookRepository(BookRepository bookRepository) {
        this.bookRepository = bookRepository;
        System.out.println("Setter Injection Done");
    }

    public void showService() {
        System.out.println("Book Service is working.");
        bookRepository.showRepository();
    }

}

```

ApplicationContext.xml:

```

<?xml version="1.0" encoding="UTF-8"?>

<beans xmlns="http://www.springframework.org/schema/beans"
       xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"

       xsi:schemaLocation="
       http://www.springframework.org/schema/beans
       https://www.springframework.org/schema/beans/spring-beans.xsd">

```

```
<bean id="bookRepository"
      class="com.library.repository.BookRepository"/>

<bean id="bookService"
      class="com.library.service.BookService">

    <!-- Constructor Injection -->
    <constructor-arg ref="bookRepository"/>

    <!-- Setter Injection -->
    <property name="bookRepository"
              ref="bookRepository"/>

</bean>
</beans>
```

Main.java

```
package com.library;

import com.library.service.BookService;
import org.springframework.context.ApplicationContext;
import org.springframework.context.support.ClassPathXmlApplicationContext;

public class Main {

    public static void main(String[] args) {

        ApplicationContext ctx =
            new ClassPathXmlApplicationContext("applicationContext.xml");

        BookService service =
            (BookService) ctx.getBean("bookService");

        service.showService();
    }
}
```

OUTPUT:

Run

Main x



"C:\Program Files\Java\jdk-25.0.2\bin\java.exe" ...

Constructor Injection Done

Setter Injection Done

Book Service is working.

Book Repository is working.

Process finished with exit code 0